Topic 2.4 - Exponential Function Manipulation

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________ Class: ____________ Date: ____________

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Rewrite exponential expressions using exponent properties and common bases.
- Translate growth factors across equivalent input intervals using rational exponents.
- Explain how algebraic equivalence corresponds to shifts, stretches, and unchanged graphs.

Section A: Foundation and representation

Question 1

Core skill | No calculator

Simplify to the form $A(3)^x$: $18(3)^{x-4}$.

Question 2

Core skill | No calculator

Rewrite $6(16)^{x+1}$ using base 2 with a single exponent.

Question 3

Core skill | No calculator

Show that $4(9)^{x/2}$ and $4(3)^x$ are equivalent for all real x.

Question 4

Core skill | No calculator

Rewrite $\frac{5^{2x+3}}{25^{x-1}}$ as a constant.

Question 5

Core skill | No calculator

Find A and c so that $12(8)^{x-2} = A(c)^x$ with c an integer and no horizontal shift.

Question 6

Equivalent intervals | No calculator

A quantity doubles every 9 hours. Write equivalent models using (a) hours and (b) three-hour intervals.

Section B: Application, comparison, and error analysis

Question 7

Equivalent intervals | Calculator permitted

A yearly decay factor is 0.78. Determine the equivalent monthly factor and monthly percent decrease.

Question 8

Modeling | Calculator permitted

An investment grows by 1.5% per quarter. Find the effective annual growth factor and annual percent increase.

Question 9

Core skill | No calculator

Rewrite $f(x) = 10(2)^{x+4} - 7$ in the form $A(2)^x - 7$, and identify the anchor point in the shifted form.

Question 10

Error analysis | No calculator

A student rewrites $(3^x)^2$ as 3^{x+2}. Correct the error and give a numerical check at $x = 1$.

Question 11

Identity | No calculator

Find all real p such that $4(2)^{px} = 4(8)^{x}$ for every real x.

Question 12

Interpretation | Calculator permitted

Two forms of a model are $P(t) = 900(1.06)^{t}$ and $P(t) = 900(1.7908477)^{t/10}$. Explain why they are approximately equivalent and what each form emphasizes.

Section C: AP-style multiple choice

MCQ 1

No calculator

Which expression is equivalent to $7(5)^{x-2}$?

A. $175(5)^{x}$

B. $\frac{7}{25}(5)^{x}$

C. $7(5)^{x} - 2$

D. $\frac{7}{5}(5)^{x}$

MCQ 2

No calculator

A factor of 1.21 applies every two months. What is the monthly factor?

A. 1.105

B. 1.10

C. 1.21^{2}

D. 0.11

MCQ 3

No calculator

Which equality is valid for every real x?

A. $2^{x+4} = 2^x + 16$

B. $2^{4x} = 8^x$

C. $9^{x/2} = 3^x$

D. $(5^x)^3 = 5^{x+3}$

MCQ 4

No calculator

The form $A(4)^x + 3$ is equivalent to $12(4)^{x-1} + 3$. What is A?

A. 48

B. 12

C. 4

D. 3

Section D: Free-response reasoning

FRQ 1 - Equivalent forms and interpretation

Calculator permitted | 6 points

A model is written as $P(t) = 2400(1.08)^{t/3}$, where t is measured in months. (a) Identify the factor for a three-month interval. (b) Rewrite the model using a one-month factor. (c) Find the effective annual factor and annual percent increase. (d) Explain what information is clearer in each form.

FRQ 2 - Algebra and graph structure

No calculator **6 points**

Consider $f(x) = 18(3)^{x-2} - 5$. (a) Rewrite f in the form $A(3)^x - 5$. (b) Give the anchor point and horizontal asymptote from the original form. (c) Explain why the two forms have identical graphs. (d) Find $f(0)$ and solve $f(x) = 157$ exactly.